

Detecting Human-Written and AI-Generated Phishing Emails with DistilBERT and Explainable AI

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Abstract—Phishing remains a major cybersecurity threat, and the rise of large language models (LLMs) has enabled attackers to generate convincing, grammatically fluent phishing emails that evade traditional filters. Existing research has primarily addressed binary detection (phishing vs. legitimate), often relying on either classical machine learning or transformers in isolation, with limited focus on AI-generated phishing and interpretability. This paper presents a transformer-based framework for tri-class classification, distinguishing between legitimate emails, human-written phishing, and AI-generated phishing. We fine-tune DistilBERT on a hybrid dataset that combines legitimate, human-phishing, and synthetic AI-phishing emails to evaluate its detection capability under realistic conditions. To enhance transparency, we integrate explainability methods (LIME and SHAP), providing insight into token-level model decisions and highlighting distinct linguistic cues across the three classes. Experimental results show that DistilBERT achieves near-perfect accuracy ($\approx 99 - 100\%$) in all categories, with strong precision and recall for the harder-to-detect AI-generated phishing class. The contributions of this work are threefold: (1) demonstrating the effectiveness of transformer based model for phishing detection in a tri-class classification setting (legitimate emails, human-written phishing, and AI-generated phishing), (2) establishing an interpretable and reproducible evaluation protocol for future benchmarking, and (3) incorporating explainable AI (XAI) techniques, such as SHAP and LIME, to provide transparency into model predictions and improve trustworthiness.

Index Terms—phishing detection, generative AI, transformer models, DistilBERT, explainable AI, cybersecurity, email security, multi-class classification

I. INTRODUCTION

Phishing continues to be one of the most damaging forms of cybercrime, exploiting human trust rather than technical vulnerabilities [1]. Traditionally, spam and phishing were often conflated; however, they serve distinct purposes. Ham refers to legitimate email messages, spam denotes unsolicited bulk mail such as advertisements, while phishing is a malicious subset of spam engineered to deceive users into disclosing sensitive information. Distinguishing among these categories is operationally important: spam is typically a nuisance, but phishing frequently results in financial and reputational harm [2].

The recent rise of generative AI has intensified this problem. Large language models (LLMs) can produce grammatically polished and contextually convincing phishing messages that evade both human judgment and legacy spam filters. Unlike early phishing attempts, which were often riddled with errors, AI-generated messages display higher fluency and persuasiveness, significantly increasing their success rate [2], [3]. Cybersecurity agencies and recent studies warn that such advances create an expanded attack surface that requires updated detection strategies [4].

Most academic work still frames phishing detection as a binary classification task. However, a three-way classification that separates legitimate emails, human-written phishing, and AI-generated phishing provides greater practical value. In real-world deployments, this tri-class setup enhances situational awareness: security analysts can distinguish between nuisance spam, traditional phishing campaigns, and highly sophisticated AI-driven attacks. Such granularity supports adaptive filtering policies, targeted user training, and more accurate threat attribution, capabilities that binary systems cannot provide [3], [5].

To address this, we propose a detection framework based on DistilBERT, a distilled version of BERT that achieves approximately 97% of BERT's performance while being 40% smaller and 60% faster [6]. This balance of efficiency and accuracy makes it suitable for deployment in real-time filtering. Combined with explainable AI (XAI) techniques such as LIME and SHAP [7], our model offers robust classification and interpretability, thus improving both performance and analyst trust.

The remainder of this paper is organized as follows: Section II reviews the related literature, Section III presents the methodology, Section IV discusses the results and findings, and Section V concludes with future directions.

II. RELATED LITERATURE

A. Background

Phishing detection has traditionally been framed as a binary classification task, distinguishing legitimate emails (“ham”) from phishing attempts. Early studies relied on stylometric and lexical features, such as n-gram distributions, word frequency, readability indices, and part-of-speech tagging, which were then fed into classical classifiers such as Naive Bayes, Logistic Regression, Support Vector Machines (SVMs) and Random Forests [1], [2]. Although these models offered reasonable performance, they lacked robustness against more sophisticated phishing strategies.

B. Literature Background

Eze and Shamir [4] introduced a publicly available corpus of AI-generated phishing emails, produced using the DeepAI API, and benchmarked several machine learning pipelines including MALLET, UDAT, LSTM-based models, and ensemble techniques. Their results suggested that AI-generated phishing displays unique stylometric patterns, such as longer word length, higher pronoun/verb frequency, and greater lexical diversity. However, their work did not investigate transformer-based approaches or interpretability. Jamal and Wimmer [5] explored transformer architectures for multi-class classification (ham, spam, phishing). Their findings showed that transformers can feasibly separate categories beyond binary labels, but they did not address AI-generated phishing or integrate explainability.

Meléndez et al. [6] compared classical classifiers (Logistic Regression, Random Forests, Naïve Bayes, SVM) with transformer models (DistilBERT, BERT, RoBERTa) for binary detection. RoBERTa achieved the highest accuracy (99.4%), slightly surpassing SVM (98.7%), reaffirming that transformers generally outperform feature-driven methods. Still, their study was limited to binary classification. Explainability in phishing detection has been addressed only sporadically. Uddin and Sarker [7] applied LIME on DistilBERT to produce token-level justifications in a binary setting. Kuikel et al. [9] advanced this line by introducing SHAP-based consistency checks for evaluating explanation reliability. While both approaches improved interpretability, they were restricted to binary datasets such as Enron and Nazario and did not explore tri-class detection.

In general, the literature demonstrates a trajectory from classical ML baselines transformer-based architectures early integration of XAI techniques, but a unified solution covering AI-generated phishing, triclass classification, and systematic interpretability has yet to be developed.

C. Research Gaps

Despite notable progress, several limitations remain in prior research: (1) Most studies focus on binary setups, neglecting the distinction between human-written and AI-generated phishing. (2) Triclass classification is underexplored, despite its operational benefits in identifying attackers, training users, and prioritizing defense. (3) Explainability methods (LIME,

SHAP) have been applied individually and rarely within phishing-specific frameworks. (4) Classical ML and transformers have often been compared in isolation, without systematic integration. (5) Many works rely on small or synthetic datasets, limiting reproducibility and external validity.

Research Gap Statement: There is limited work addressing tri-class classification that separates legitimate, human-written, and AI-generated phishing emails. Furthermore, systematic integration of explainable AI within transformer models remains scarce.

D. Objectives and Contributions

The objectives of this study are: (1) To develop a tri-class classification framework distinguishing legitimate, human-written phishing, and AI-generated phishing emails, thereby enhancing situational awareness. (2) To fine-tune DistilBERT as a lightweight transformer capable of delivering high accuracy in practical settings. (3) To integrate LIME and SHAP to provide interpretable insights into model predictions. (4) To evaluate the framework on a diverse dataset, ensuring reproducibility and operational value. (5) Resource for future work: A reproducible experimental pipeline to support systematic benchmarking in phishing detection research.

III. METHODOLOGY

A. Framework Overview

The proposed system employs a lightweight transformer, DistilBERT [9], augmented with explainable AI (XAI) techniques [10] to perform tri-class classification of emails. Specifically, the model discriminates among legitimate (ham), human-authored phishing, and AI-generated phishing. This combination balances efficiency, accuracy, and interpretability, making it suitable for deployment in practical security environments.

B. Dataset

The dataset combines legitimate emails [11], human phishing emails [12], and AI-generated phishing emails [4],[13]. To reflect realistic traffic, legitimate emails were slightly overrepresented compared to phishing instances. Table II shows the distribution.

C. Preprocessing and Feature Representation

For transformer training, email bodies were directly tokenized using DistilBERT’s WordPiece tokenizer, which breaks text into subword units while preserving semantic context. Unlike feature-engineered ML pipelines that rely on lexical indicators (e.g., n-grams, word counts), this approach produces dense embeddings optimized for contextual representation. This ensures end-to-end learning without manual feature design.

TABLE I
SUMMARY OF KEY RELATED WORK ON PHISHING DETECTION

Study	Task	Model(s)	Dataset(s)	Explainability	Key Findings
Eze & Shamir (2024) [4]	Human vs. AI phishing	ML ensembles	AI-phish corpus	✗	AI phishing shows distinct lexical cues.
Jamal & Wimmer (2023) [5]	Multi-class (ham/spam/phish)	BERT variants	Custom dataset	✗	Showed feasibility of multi-class classification.
Meléndez et al. (2024) [6]	Binary (ham vs. phishing)	SVM, RF, BERT, RoBERTa	Enron + others	✗	Transformers outperform ML.
Uddin & Sarker (2024) [7]	Binary (phish vs. ham)	DistilBERT + LIME	Enron, Nazario	✓	Token-level rationales improve trust.
Kuikel et al. (2025) [8]	Binary phishing detection	BERT, LLaMA, Wizard	Enron, Nazario	✓	Proposed CC-SHAP for consistent explanations.

TABLE II
DATASET DISTRIBUTION

Class	Count	% of Total
Legitimate	3000	54%
Human Phishing	1565	28%
AI Phishing	1069	19%
Total	5634	100%

D. Experimental Setup

Experiments were executed in Google Colab, leveraging GPU acceleration. Models were implemented with PyTorch and Hugging Face Transformers. DistilBERT was fine-tuned using the Trainer API, optimized with AdamW and learning rate scheduling. Early stopping prevented overfitting, with training converging within three epochs.

E. Evaluation Protocol

Performance was evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. A stratified train-validation-test split ensured balanced class representation. Confusion matrices were also generated to identify common misclassifications across the three categories.

- Accuracy : overall correctness (but may mislead under class imbalance).
- Precision : likelihood that detected phishing is truly phishing.
- Recall : proportion of phishing correctly captured.
- F1-score : harmonic mean of precision and recall.
- Macro-average : treats each class equally.
- Weighted-average : accounts for class imbalance.

F. Explainability Layer

To ensure model transparency, the framework integrates two complementary XAI tools:

- 1) LIME (Local Interpretable Model-Agnostic Explanations) : Highlights tokens that most influence classification decisions at the local (per-email) level.
- 2) SHAP (SHapley Additive exPlanations) : Provides global importance scores across the dataset and token-level heatmaps.

These explanations clarify why an email is flagged, making the model interpretable to analysts and increasing user trust.

IV. RESULTS AND DISCUSSION

A. Model Performance

The fine-tuned DistilBERT model achieved consistently high performance across all three classes. Table III reports precision, recall, and F1-scores, indicating near-perfect classification. Notably, AI-generated phishing was detected with perfect precision and recall, while legitimate and human-authored phishing emails exhibited minimal error. Table III summarizes results.

TABLE III
MODEL PERFORMANCE OF DISTILBERT

Class	Precision	Recall	F1-score
Legitimate	1.00	0.99	1.00
Human Phishing	0.99	1.00	0.99
AI Phishing	1.00	1.00	1.00

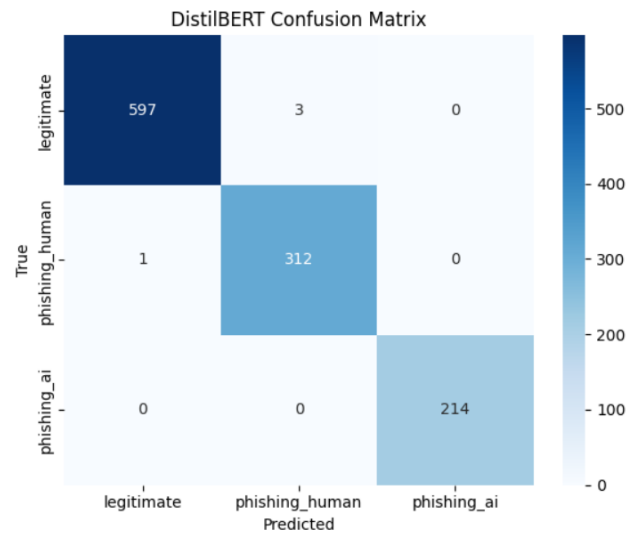


Fig. 1. Confusion matrix heatmap showing DistilBERT's tri-class classification performance.

B. Performance Interpretation

The results demonstrate DistilBERT's capacity to capture both stylistic and semantic differences in phishing content. While human-authored phishing often contained urgency

markers and grammatical inconsistencies, AI-generated phishing exhibited polished syntax and corporate style phrasing. DistilBERT distinguished these nuances while maintaining high recall on legitimate emails, an operational advantage for reducing both false positives and false negatives in real-world deployments.

C. Explainability with LIME and SHAP

To enhance transparency, we employed both local and global interpretability techniques. Local explanations with LIME provided token-level rationales for individual predictions. Tokens such as “password reset,” “verify account,” and “click link” consistently appeared as strong phishing indicators, while professional greetings and contextual references favored legitimate classifications. In borderline cases, the model did not exhibit undue confidence. For instance, (Fig. 2) when presented with an invitation email (“Here it is! Your Enron Net Works Holiday Party Invitation. Please click on the attached link to launch.”), the model predicted *phishing_ai* with only 35.1% probability, while legitimate and human-phishing classes received 32.3% and 32.6%, respectively. LIME attributed suspicion primarily to the tokens “click,” “attached,” and “launch.” These narrow probability margins highlight model uncertainty and demonstrate how interpretability guides analysts in reviewing ambiguous cases rather than relying solely on automated classification. Moreover, the plain-language summary and recommendations generated from these explanations make the output understandable to non-expert end users, enabling them to take informed action without requiring deep technical knowledge.

Global explanations with SHAP (Fig. 3) revealed consistent linguistic trends across the dataset. AI-generated phishing tended to use polished syntax and generic corporate phrasing, whereas human-crafted phishing was characterized by urgency markers and grammatical inconsistencies. These patterns indicate that the model relies on semantically meaningful cues rather than superficial correlations, thereby increasing analyst trust in real-world deployments.

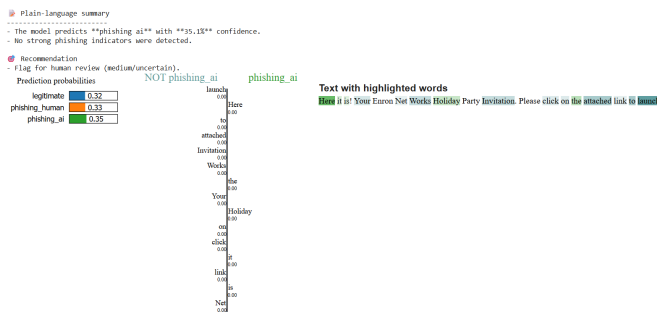


Fig. 2. Local interpretability with LIME on an ambiguous email. The model predicted *phishing_AI* with 35% confidence, where weak contributions from tokens like “click” and “link” indicate borderline classification.

The tabular summary (Table IV) highlights how tokens like “Support” and “Customer” influence human phishing classification, while tokens like “Microsoft” and “Thank” emerge as distinctive signals in AI-generated phishing. Global

TABLE IV
TOP SHAP TOKEN CONTRIBUTIONS ACROSS CLASSES. POSITIVE VALUES INDICATE STRONG ASSOCIATION.

Token	Legit.	Phish_Human	Phish_AI
Captain	0.0326	0.0214	0.0107
Team	0.0273	0.0154	0.0083
Support	0.0227	0.0268	—
Customer	0.0115	0.0272	—
Microsoft	—	—	0.0081
Thank	—	—	0.0085

interpretability revealed that AI-generated phishing often contained polished syntax and generic corporate phrasing, whereas human-authored phishing emphasized urgency and grammatical inconsistencies. These insights confirm that DistilBERT leverages meaningful linguistic cues rather than spurious correlations.

Operational Relevance: For SOC analysts, these interpretability layers strengthen analyst confidence, provide forensic insight, and enable training programs tailored to attacker sophistication.

D. Strengths and Limitations

This study demonstrates three main strengths:

- 1) High accuracy in tri-class classification (legitimate, human phishing, AI-phishing).
- 2) Use of DistilBERT, balancing efficiency and accuracy.
- 3) Systematic integration of explainability tools (LIME and SHAP).

However, limitations remain. The dataset size, though balanced, is modest relative to real-world traffic. AI-generated phishing samples were derived from specific models, potentially underrepresenting adversarial diversity. Furthermore, although DistilBERT is lightweight compared to BERT, fine-tuning still requires GPU resources, and LIME/SHAP introduce overhead that may constrain real-time deployment.

V. CONCLUSION AND FUTURE WORK

This paper presented a DistilBERT based framework for tri-class phishing detection, distinguishing legitimate emails, human-written phishing, and AI-generated phishing. By integrating explainability techniques (LIME and SHAP), the model achieved both high classification accuracy and transparent decision making, supporting trust and usability in security operations.

The framework’s main strengths are its ability to separate AI-generated from human phishing, its computational efficiency relative to full BERT, and its interpretable outputs. In practice, the tri-class setup provides greater situational awareness for security analysts, enabling them to differentiate between nuisance spam, conventional phishing campaigns, and advanced AI-driven threats, an advantage that binary systems cannot provide. Limitations include the modest dataset scale and reliance on specific AI-generated phishing sources, which may not capture the full diversity of adversarial strategies.

Future work should expand datasets with more varied AI-generated samples, optimize XAI methods for real-time use,

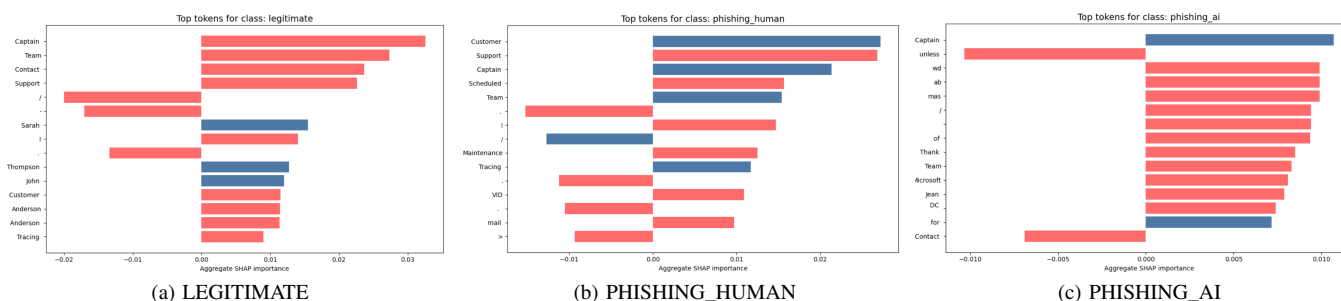


Fig. 3. Global SHAP explanations across classes. DistilBERT leverages semantically meaningful cues such as “Captain” and “Team” in legitimate emails, urgency markers in human phishing, and polished corporate phrasing in AI-generated phishing.

and explore cross-lingual detection. These directions will further enhance the robustness and practical deployment of explainable phishing defenses.

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